Functionally Decoupled Visual Intent Learning under Supervisory and Semantic Ambiguity

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Abstract

Visual intention understanding aims to infer the latent purpose conveyed by an image, and is fundamentally more challenging than conventional image classification due to its abstract and subjective nature. A central difficulty of this task lies in two complementary forms of ambiguity: supervisory ambiguity, caused by subjective annotation over plausible intentions, and semantic ambiguity, caused by semantically adjacent and overlapping intents. In this work, we propose FDIL (**F**unctionally **D**ecoupled **I**ntent **L**earning), a framework that treats these two sources of ambiguity through complementary roles. FDIL first introduces a semantic prior built from textual cues to provide structured candidate semantics, and then learns image-aware intent prediction under uncertainty-guided teacher supervision. Concretely, we develop Semantic Local Reranking and Calibration (SLR-C) to form the semantic prior, and Uncertainty-guided Text Distillation (UTD) to transform annotator disagreement into an adaptive training signal. By combining semantic structure with uncertainty-aware learning, FDIL improves both the stability of supervision and the discrimination of semantically similar intents. On Intentonomy, FDIL leads the state-of-the-art by 6.1% relative average F1 and 28.0% on the hard subset. Extensive experiments demonstrate consistent gains in both hard-sample robustness and overall multi-label prediction performance. When transferred to the Flickr-LDL and Emotion6 label-distribution benchmarks, FDIL re-

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mains effective and the division of labour between its two modules persists. Overall, FDIL provides a functionally decoupled perspective for visual understanding and we hope this approach will inspire further research in the field.
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1. Introduction

With the rapid proliferation of social media, images are increasingly used as a medium for users to share daily experiences, emotions, and personal viewpoints. Visual intention recognition aims to infer the latent purpose conveyed by an image, and has become an important problem in human-centered multimedia analysis [1, 2]. Accurately recognizing these intentions is valuable for a variety of downstream applications, including user interest understanding, psychological and behavioral profiling, broader social media content analysis, *etc*.

Compared with conventional image classification, visual intention recognition is fundamentally more challenging. In traditional classification tasks, the goal is to assign images to relatively objective semantic categories, such as objects, scenes, or actions. In contrast, intention recognition requires mapping visual evidence to abstract and subjective intent labels. These labels are rarely determined by local visual cues alone. Instead, they often emerge from a holistic interpretation of scene context, human activities, object configurations, social interactions, and commonsense knowledge. Therefore, the main difficulty of this task lies not merely in visual representation learning, but in establishing a reliable bridge between concrete image content and abstract intention semantics.

A central challenge of this task is its inherent ambiguity, which manifests in two distinct forms, as shown in Figure 1 (a). The first is semantic ambiguity. Intent labels are often semantically adjacent, partially overlapping, or hierarchically correlated. Different labels may correspond to visually similar patterns, while the same intent may be expressed through diverse appearances. This leads to high inter-class similarity and substantial intra-class variation, making fine-grained discrimination particularly diffi-

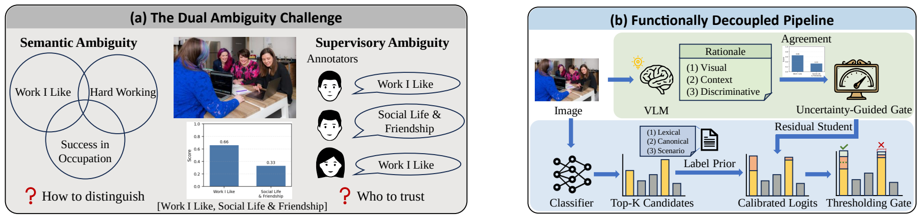

Figure 1: **Functionally decoupled ambiguity modeling in FDIL.** (a) Visual intent recognition suffers from semantic ambiguity among overlapping intents and supervisory ambiguity from annotator disagreement. (b) FDIL separates semantic refinement via Semantic Local Reranking and Calibration (SLR-C) from uncertainty-aware semantic supervision via Uncertainty-guided Text Distillation (UTD).

cult. The second is supervisory ambiguity. Since intention recognition is intrinsically subjective, different audiences may infer different yet plausible intentions from the same image. As a result, the supervisory signal itself may become unstable, especially for difficult samples with severe disagreement. Under such circumstances, directly optimizing against hard labels can force the model to fit oversimplified or unreliable targets, thereby harming the robustness of the learned decision boundaries. Recent studies have increasingly argued that such disagreement should not be treated as mere annotation noise, but as meaningful uncertainty that reflects the structure of subjective interpretation itself [3, 4].

Existing studies have attempted to alleviate these difficulties from different perspectives. Prototype-based methods, *e.g.*, PIP-Net [5], introduce representative anchors to reduce feature-space confusion. Structure-aware methods, including CPAD [6] and HLEG [7], exploit hierarchical label relationships to enhance multi-level semantic modeling and strengthen supervision. From the perspective of label quality, LabCR [8] explicitly models label shifting and label blemish to better cope with imperfect annotation distributions. More recently, knowledge-enhanced approaches have recognized that visual evidence alone is often insufficient for subjective intention understanding. IntCLIP [9] incorporates the vision-language alignment knowledge of CLIP into the task, HVU-CLIP [10] refines intent semantics iteratively and couples them with bidirectional visual-semantic interaction. IntentMLM [11] further leverages multimodal large models, caption generation, and retrieval to perform more holistic intention perception , while CoT4Intent [12] attributes the residual errors of MLLM-based intent classification to hallucination and suppresses it with structured chain-of-thought reasoning. This line of work reflects a broader recent shift toward large vision-language and multimodal large language models (MLLMs) as sources of external semantic knowledge for subjective image understanding. However, our zero-shot experiments show that prompting a general-purpose MLLM directly remains below a task-trained model on this benchmark. Such models supply useful intent priors but do not provide the dataset-specific calibration and uncertainty-aware supervision that subjective intent recognition requires. Our purpose is therefore not to replace these models, but to convert their semantic knowledge into a structured prior and an uncertainty-gated training signal that explicitly target the two ambiguities below.

Although these methods have yielded encouraging progress, they typically treat one perspective of ambiguity as dominant. In our view, the core difficulty of subjective visual intention recognition is not any single ambiguity alone, but the coexistence of supervisory ambiguity and semantic ambiguity, which interfere with different functional components of the system. The former primarily corrupts supervision by compressing diverse subjective interpretations into rigid targets, whereas the latter primarily distorts local decision geometry by making neighboring labels difficult to separate. Treating these two failure modes within one mechanism may therefore be suboptimal, as it is difficult to simultaneously achieve robust learning on disagreement-prone samples and precise discrimination among closely related intents.

Motivated by this observation, we propose the **FDIL** (**F**unctionally **D**ecoupled **I**ntent **L**earning), a functionally decoupled framework for subjective intent recognition, as illustrated in Figure 1(b), where fixed semantic priors provide structured candidate semantics, uncertainty-aware text teachers stabilize ambiguous supervision, and a residual student learns image-aware intent prediction on top of the semantic prior. We decouple ambiguity handling into two complementary functions. First, Semantic Local Reranking and Calibration (SLR-C) provides ambiguity-aware semantic refinement, where heterogeneous lexical, canonical, and scenario priors define a semantic prior that supplies structured candidate semantics. Second, Uncertainty-guided Text Distillation (UTD) provides ambiguity-aware semantic supervision, where a text-only teacher distilled from rationale embeddings regularizes the visual model according to annotator agreement, improving robustness under subjective supervision and supporting image-aware prediction on top of the semantic prior. Together, these two functions improve both robustness and discriminability without introducing high architectural complexity.

Extensive experiments on the Intentonomy dataset show that our FDIL outperforms baselines and recent state-of-the-art methods, yielding improvements in both hard-sample robustness and overall multi-label prediction quality. We further transfer FDIL to the Flickr-LDL and Emotion6 label-distribution benchmarks, where the division of labour survives the change of prediction space.

The main contributions of this work are summarized as follows:

- We reformulate visual intention recognition as a *functionally decoupled* ambiguity problem, treating supervisory ambiguity and semantic ambiguity as two separately justified functional roles rather than a single undifferentiated mechanism. This decomposition outperforms a strong unified alternative while keeping each function individually interpretable, separately measurable, and independently deployable.

- We propose Semantic Local Reranking and Calibration (SLR-C) together with residual semantic refinement, where heterogeneous textual priors first define a semantic prior with structured candidate semantics and a lightweight residual student further adapts prediction to image evidence.

- We propose Uncertainty-guided Text Distillation (UTD), which leverages annotator disagreement as an uncertainty signal to adaptively introduce semantic guidance, improving robustness under ambiguous supervision.

2. Related Work

Early visual intent studies target specific domains such as persuasive or political imagery [13, 14, 15], while the Intentonomy benchmark [1] enables general-purpose intent understanding in unconstrained social media. Recent Intentonomy methods address the task from complementary angles. PIP-Net [5] uses prototype-based representations for interpretable intent perception, CPAD [6] and HLEG [7] exploit cross-modality or hierarchical label structures, LabCR [8] models label calibration and relation preservation, and IntCLIP [9] and IntentMLM [11] introduce CLIP/MLLM-based semantic knowledge. More recently, HVU-CLIP [10] extends IntCLIP toward holistic visual understanding through iterative semantic label refinement and bidirectional visual-semantic interaction, while CoT4Intent [12] investigates hallucination in MLLM-based intention classification and mitigates it through structured chain-of-thought reasoning. Robust-loss methods such as Focal Loss [16], OHEM [17], collaborative refining [18], and ASL [19] reduce the effect of difficult, noisy, or imbalanced samples through reweighting or selective optimization. They treat ambiguity mainly as an optimization issue, however, rather than as a subjective semantic signal. Disagreement-aware learning instead preserves label diversity. Peterson et al. [3] train on human label distributions, Davani et al. [4] model annotator-specific labels, DisCo [20] jointly models item- and annotator-level distributions, and CrowdOpinion [21] recovers minority opinions. Label distribution learning (LDL) takes the complementary route of predicting the full annotation distribution, where LRR [22] and DPA [23] preserve label ranking relations, δ-LDL [24] relaxes exact distribution recovery to approximate correctness, and LDL-DVS [25] keeps the divisiveness of conflicting labels consistent. These objectives regularize the output distribution but leave the semantics of the label space and the reliability of each supervision signal untouched. In contrast, FDIL explicitly treats annotator disagreement as an uncertainty signal and adaptively regulates semantic guidance according to supervision ambiguity, yielding more robust learning under subjective supervision.

3. Method

3.1. Overview

To address the dual challenges of supervisory ambiguity and semantic ambiguity in visual intention recognition, we propose the **FDIL**, a functionally decoupled frame-

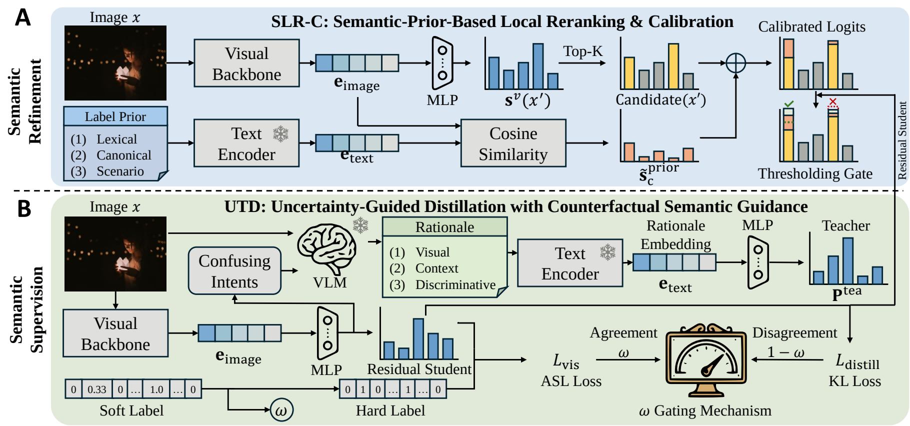

Figure 2: Overview of FDIL. SLR-C constructs a textual semantic prior and performs local reranking plus residual refinement. UTD uses a rationale-based text teacher and an agreement-aware gate to stabilize learning under ambiguous supervision.

work where fixed semantic priors provide structured candidate semantics, uncertainty-aware text teachers stabilize ambiguous supervision, and a residual student learns image-aware intent prediction on top of the semantic prior. Instead of forcing one unified mechanism to handle both supervisory ambiguity and semantic ambiguity, we assign them to two complementary functions.

FDIL consists of two key components, as shown in Figure 2.

- **Semantic Local Reranking and Calibration (SLR-C)**, which constructs a semantic prior from heterogeneous textual cues and supports image-aware refinement through a residual student.

- **Uncertainty-guided Text Distillation (UTD)**, which introduces adaptive semantic guidance to improve robustness under ambiguous supervision.

Given an input image, the model first produces visual intent scores using a strong visual backbone. SLR-C then provides structured candidate semantics by locally reranking intent candidates and forming a semantic prior. UTD supplies uncertainty-aware semantic supervision through a rationale-based text teacher and can be reused to regularize both the visual student and the residual refinement module built on top of the semantic prior.

3.2. Problem Formulation

Given an input image x, the goal is to predict a set of intent labels from a predefined label space $\mathcal{Y} = \{1, \ldots, C\}$, where C is the total number of intent categories. Due to the subjective nature of visual intention recognition, annotations are provided as soft label vectors $\mathbf{y}^{\text{soft}} \in \{0, 0.33, 0.66, 1\}^C$, reflecting annotator agreement.

Existing approaches typically follow either hard binarization or soft regression paradigms, both of which have limitations under ambiguous supervision. To address this, we adopt a **decoupled utilization** of annotation signals.

Specifically, we derive binarized labels $\mathbf{y}^{\text{hard}}$ for classification supervision, while extracting an image-level agreement score $\omega \in (0, 1]$ from $\mathbf{y}^{\text{soft}}$. Here, ω reflects annotator consensus, where higher values indicate more reliable supervision. The binarized labels are computed as.

$$y_c^{\text{hard}} = \begin{cases} 1, & \text{if } y_c^{\text{soft}} > 0 \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

In practice, an image may contain multiple positive intents with different agreement levels. We therefore adopt a conservative pooling rule and define the sample-level agreement as the minimum soft agreement among all positive labels.

$$\omega = \min_{c:\, y_c^{\text{hard}}=1} y_c^{\text{soft}}. \tag{2}$$

This choice lets the hardest positive label determine how much the sample should trust direct supervision.

3.3. Semantic Local Reranking and Calibration (SLR-C)

Semantic ambiguity mainly manifests as local confusion among visually plausible but semantically adjacent intents. A candidate-recall diagnostic shows that the baseline Top-10 candidates already cover 83.9% of positive labels, and 67.8% of its false negatives remain recoverable within this set. This suggests that many errors arise from ranking confusable intents rather than failing to retrieve the correct intent. To address this, we build a semantic refinement module that provides structured candidate semantics before turning to ambiguity-aware supervision.

3.3.1. Semantic Prior Construction

Intent semantics are inherently multi-granular and abstract, making a single textual representation insufficient. We therefore construct a heterogeneous semantic prior for each intent by combining three complementary textual sources, namely lexical intent phrases, canonical intent descriptions, and descriptive visual scenarios. The lexical source provides concise intent name cues, the canonical source provides semantic definitions, and the scenario source grounds abstract intents into plausible visual situations.

Example. For the intent *SocialLifeFriendship*, the lexical phrase is *"Social Life and Friendship"*, and the canonical description is. *"Being part of a social group. Having people to do things with. Having close friends, others to rely on. Making friends, drawing others near."* This intent is further expanded into multiple scenario archetypes with distinct visual realizations. For *SocialLifeFriendship*, scenarios include friends hiking together, close friends talking on a couch, a group dinner, or volunteers gardening together.

The scenarios are not hand-picked but produced by a fixed generation protocol applied uniformly to every intent class (template released with our artifacts), under which the vision-language model enumerates visually distinct realizations along five axes (subjects/objects, actions, settings, atmosphere, and photographic style), with at least three settings per intent so as to span intra-class variation rather than a single canonical view. Inclusion favors representativeness over exhaustive coverage. Outside the generated scenarios the prior degrades gracefully toward the lexical and canonical sources, the base logits, and the residual student.

Given an image x, we score each intent by the CLIP-scaled cosine similarity between the image feature $f_{\text{img}}(x)$ and the text embedding $\mathbf{e}_c$ of class c.

$$s_c^{\text{prior}} = \text{sim}(f_{\text{img}}(x), \mathbf{e}_c) = \gamma \cdot \frac{f_{\text{img}}(x)^\top \mathbf{e}_c}{\|f_{\text{img}}(x)\|_2 \|\mathbf{e}_c\|_2}, \tag{3}$$

where γ is the CLIP logit scale. Each source $r \in \{\text{lex}, \text{can}, \text{scen}\}$ is normalized independently across the C classes and the three are averaged into the prior.

$$\tilde{s}_c^r = \frac{s_c^r - \mu^r(x)}{\sigma^r(x) + \epsilon}, \qquad \tilde{s}_c^{\text{prior}} = \frac{1}{3} \sum_r \tilde{s}_c^r, \tag{4}$$

where $\mu^r(x)$ and $\sigma^r(x)$ are the per-sample mean and standard deviation over the C intent classes and ϵ ensures numerical stability.

3.3.2. Semantic Local Reranking

In the refinement module, we start from base visual logits $\mathbf{s}^b$ produced by a frozen visual student. We first select a compact candidate set as.

$$C_K(x) = \mathrm{TopK}(\mathbf{s}^b, K). \tag{5}$$

The base logits are then locally adjusted with the normalized prior.

$$\tilde{s}_c = \begin{cases} s_c^b + \alpha \cdot \tilde{s}_c^{\mathrm{prior}}, & c \in C_K(x), \\ s_c^b, & \text{otherwise}. \end{cases} \tag{6}$$

Here α controls the strength of semantic reranking. In our implementation, α is fixed to 0.3.

This localized reranking produces a semantic prior that narrows the candidate set and sharpens the local score geometry without introducing an additional trainable prior module.

3.3.3. Residual Refinement on the Semantic Prior

Fixed reranking alone is not sufficient, because the heterogeneous prior may still fall short of the image-specific evidence. We therefore train a lightweight residual student on top of the semantic prior.

$$s_c^{\mathrm{full}} = \tilde{s}_c + h_{\mathrm{res}}(f_{\mathrm{img}}(x))_c, \tag{7}$$

where $h_{\mathrm{res}}(\cdot)$ is a residual MLP that predicts class-wise residual intent scores from the image feature. In practice, the SLR-C logits $\tilde{s}_c$ are first computed and frozen, and the residual module is then optimized separately.

In the final full model, this residual module is not trained in isolation, since it can also be regularized by the uncertainty-aware text teacher introduced next. Thus, SLR-C provides the structured candidate semantics first, and UTD later supplies the ambiguity-aware supervision used to stabilize image-aware prediction on top of the semantic prior.

3.3.4. Class-wise Calibration

Motivated by Guo et al. [26], to account for class imbalance and varying score distributions, we adopt class-wise thresholds.

$$\hat{y}_c = \mathbb{I}[\sigma(s_c^{\text{full}}) > t_c], \tag{8}$$

where t_c is determined on the validation set. Concretely, each threshold is selected independently by maximizing the class-wise validation F1 score.

$$t_c^* = \arg\max_{t \in \mathcal{T}} \mathrm{F1}_c^{\text{val}}(t), \tag{9}$$

where $\mathcal{T}$ is a fixed threshold grid. This design lets each class adapt its decision boundary to its own score distribution and frequency level, instead of sharing a single global threshold.

3.4. Uncertainty-guided Text Distillation (UTD)

Supervisory ambiguity arises when multiple plausible intents coexist for a given image. Direct optimization against hard labels may lead to overconfident predictions and unstable decision boundaries. To alleviate this issue, we introduce semantic guidance from text space and adapt its influence based on the uncertainty of supervision.

3.4.1. Text-derived Teacher

For each training image, we construct structured textual descriptions that capture semantic cues relevant to intention understanding. These descriptions are generated offline by a vision-language model, guided by both the image content and intent semantics.

To enhance discriminative capability under semantic ambiguity, we further incorporate contrastive cues by identifying visually plausible but incorrect intents predicted by the current model. Specifically, we select high-confidence negative intents as confusion candidates and use them to guide the generation of discriminative textual descriptions. This encourages the resulting text to emphasize subtle distinctions among semantically adjacent intents.

The generated text aims to summarize three aspects. (1) observable visual elements, (2) contextual interpretation beyond direct perception, and (3) discriminative cues for distinguishing confusing intents.

The resulting text $T(x)$ is encoded into a dense embedding space using a frozen text encoder.

$$\mathbf{e}_{\text{text}} = f_{\text{text}}(T(x)). \tag{10}$$

A lightweight classifier then maps the embedding into a multi-label teacher distribution.

$$\mathbf{p}^{\text{tea}} = \sigma(g_{\text{tea}}(\mathbf{e}_{\text{text}})/\tau), \tag{11}$$

where $\sigma(\cdot)$ is the sigmoid function and τ is the temperature parameter.

This design introduces contrastive semantic signals that improve the teacher's ability to distinguish semantically similar intents, while remaining separate from the semantic prior itself. In the full FDIL pipeline, the same teacher can further regularize the residual refinement module built on top of SLR-C.

3.4.2. Uncertainty-guided Distillation

The visual backbone produces logits $\mathbf{s}^v$ for image x. We adopt the Asymmetric Loss [19] for classification.

$$\mathcal{L}_{\text{vis}} = \frac{1}{C}\sum_{c=1}^{C} \ell_{\text{ASL}}(s_c^v, y_c^{\text{hard}}). \tag{12}$$

To incorporate semantic guidance, we introduce an uncertainty-guided Bernoulli distillation loss.

$$\mathcal{L}_{\text{distill}} = \sum_{c=1}^{C} \text{KL}(\text{Bern}(p_c^{\text{tea}}) \,\|\, \text{Bern}(\sigma(s_c^v/\tau))), \tag{13}$$

which is applied independently to each label. We use Bernoulli-style distillation rather than softmax distillation because intent recognition is a multi-label problem in which each intent is modeled as an independent Bernoulli variable rather than as a mutually exclusive class on a probability simplex. This formulation preserves per-class uncertainty and avoids imposing artificial competition among labels.

Rather than mixing the two objectives with a fixed global coefficient, we use a gate derived from the agreement score. For the current sample, the teacher branch weight is defined as $g = 1 - \omega$, so low-agreement samples receive stronger semantic guidance, while high-agreement samples remain dominated by direct visual supervision.

Using g, the final loss is defined as.

$$\mathcal{L}_{\text{total}} = (1 - g)\mathcal{L}_{\text{vis}} + \lambda g \mathcal{L}_{\text{distill}} = \omega \mathcal{L}_{\text{vis}} + \lambda(1 - \omega)\mathcal{L}_{\text{distill}}, \tag{14}$$

where λ is the distillation loss weight.

This gated formulation allows the model to rely more on visual supervision when annotations are reliable, and to leverage semantic guidance when supervision is uncertain. In other words, UTD stabilizes ambiguous supervision through an uncertainty-aware text teacher rather than by modifying the semantic prior itself.

3.5. Functionally Decoupled Ambiguity Modeling

We interpret visual intention recognition as a functionally decoupled ambiguity problem, formed by semantic confusion among neighboring intents and supervisory uncertainty from subjective annotations.

Two ambiguities, two learning problems. Let X denote the image, $Z = \phi(X)$ its deterministic visual representation, and $Y_c \in \{0, 1\}$ a random human judgement for intent c. Then

$$H(Y_c \mid Z) = \underbrace{H(Y_c \mid X)}_{\text{irreducible supervisory}} + \underbrace{I(Y_c; X \mid Z)}_{\text{discarded by } \phi}, \tag{15}$$

which follows from $I(Y_c; X \mid Z) = H(Y_c \mid Z) - H(Y_c \mid X, Z)$ and $H(Y_c \mid X, Z) = H(Y_c \mid X)$ because $Z = \phi(X)$. The first term is disagreement persisting given the complete image. Being a functional of the annotation distribution $\Pr(Y_c \mid X)$ alone, it cannot be reduced by any choice of ϕ or any predictor reading Z. The second is label-relevant information that X carries but Z discards, a functional of the representation map. (15) serves to show that the two deficits are functionals of different objects and hence pose two learning problems, what the model is asked to fit and what information the decision rule can access, which is why FDIL treats them with mechanisms of different type.

SLR-C supplies label-relevant information the baseline underuses. Let $B = \mathbf{s}^b(Z)$ be the baseline logits and $S = \tilde{\mathbf{s}}^{\text{prior}}(X)$ the text-side prior scores. Since B is a function of Z, the data-processing inequality gives $H(Y_c \mid B) \geq H(Y_c \mid Z)$, showing that the baseline prediction is in general insufficient, retaining only part of the label-relevant information. Conditioning additionally on the prior gives the identity

$$H(Y_c \mid B, S) = H(Y_c \mid B) - I(Y_c; S \mid B), \tag{16}$$

so residual uncertainty falls by the conditional mutual information $I(Y_c; S \mid B)$, the label-relevant information in the prior that the baseline prediction does not already carry, which is zero precisely when S is redundant given B. SLR-C is therefore an injection of under-used text-side information, and $\tilde{s}_c = s_c^b + \alpha \tilde{s}_c^{\text{prior}}$ is the minimal way to make the decision rule a function of (B, S) rather than of B alone.

Top-K reranking as local hypothesis screening. For a positive intent c^+ and a confusing intent c^- in $C_K(x)$, let $m = s_{c^+}^b - s_{c^-}^b$ and $\Delta_{\text{prior}} = \tilde{s}_{c^+}^{\text{prior}} - \tilde{s}_{c^-}^{\text{prior}}$, and the reranking rule gives

$$m' - m = \alpha \Delta_{\text{prior}}, \tag{17}$$

so the margin grows whenever the prior ranks c^+ above c^-, a reversed pair is corrected when $\alpha \Delta_{\text{prior}} \geq -m > 0$, and scores outside $C_K(x)$ are untouched. Restricting the prior to $C_K(x)$ is thus local hypothesis screening, in the sense of variable screening for high-dimensional selection. The semantic channel acts only on hypotheses the visual evidence already deems plausible. K balances coverage against distractor noise. SLR-C therefore does not globally rescore the label space but corrects semantic confusion among high-probability candidates.

UTD as reliability-weighted privileged information. The rationale teacher reads a textual description $T(x)$ generated with access to intent semantics and model-specific confusion candidates, which exists at training time only. UTD therefore uses privileged information [27], realized through distillation [28], and must decide how much to trust that privileged channel. In a per-sample scalar view of one label, the annotated target is $y_v = y + \varepsilon_v$ with $\mathbb{E}\varepsilon_v = 0$ and $\mathrm{Var}(\varepsilon_v) = \sigma_v^2(\omega)$, decreasing in the agreement score ω since higher consensus means less annotation noise, while the teacher gives $y_t = y + b_t + \varepsilon_t$ with bias b_t and independent noise of variance σ_t^2. For $\hat{y}(g) = (1-g)y_v + g\, y_t$ we have $\mathrm{MSE}(g) = (1-g)^2\sigma_v^2(\omega) + g^2(\sigma_t^2 + b_t^2)$, minimized at

$$g^* = \frac{\sigma_v^2(\omega)}{\sigma_v^2(\omega) + \sigma_t^2 + b_t^2}, \tag{18}$$

which increases with supervision noise, decreases with teacher noise and bias, and is monotonically decreasing in ω. Our gate $g = 1 - \omega$ is a monotone, parameter-free approximation of (18). It keeps the correct direction without estimating σ_t^2 or b_t^2, and reaches $g = 0$ at $\omega = 1$, where supervision is effectively noiseless and the privileged channel can only add bias. The teacher contribution to the student-logit gradient is then

$$\frac{\partial\, \lambda(1-\omega)\mathcal{L}_{\text{distill}}}{\partial s_c^v} = \frac{\lambda(1-\omega)}{\tau}(p_c - q_c), \tag{19}$$

with $p_c = \sigma(s_c^v/\tau)$ and $q_c = p_c^{\text{tea}}$, so guidance is routed in proportion to annotator disagreement.

In summary, SLR-C addresses prediction-semantic insufficiency in the decision space, while UTD addresses unreliable supervision through uncertainty-weighted privileged information. Since each module is tied to a different deficit, their gains should be distinctly sensitive to the two ambiguity sources, which we test in Sec. 4.3.2.

4. Experiments

4.1. Experimental Setup

4.1.1. Dataset and Evaluation Metrics

We evaluate our method on the Intentonomy dataset [1], a multi-label benchmark for visual intention recognition in social media scenarios. Following the standard split used in our experiments, the dataset contains 12,739 training images, 498 validation images, and 1,216 test images. Intentonomy additionally provides soft agreement annotations with values in $\{0, 0.33, 0.66, 1\}$ on the training set, which directly reflect the degree of annotator consensus and serve as the uncertainty signal ω in our framework.

We use Macro F1, Micro F1, Samples F1 as the evaluation metrics. To be consistent with recent Intentonomy works, we additionally report *Avg.*, which is the average of three F1 scores. Unless otherwise stated, all reported numbers are percentages and all results are computed on the Intentonomy test split.

To better understand the effectiveness of our method, we further conduct experiments on subsets of the intent classes. We follow the work of Jia et al. [1] and break down intent classes according to content difficulty and dependency. Difficulty classes are measured by how much a standard ResNet-50 model outperforms the random results. Content classes are divided into object-dependent (*Object*-classes), context-dependent (*Context*-classes), and *Others*, which depend on both foreground and background information. We emphasize the *Hard* score throughout the analysis, since it measures the ambiguity-prone categories where semantic overlap and annotator disagreement are most severe.

For the cross-dataset study we use two label-distribution benchmarks, Flickr-LDL [29] (11,150 Flickr images, eight emotions) and Emotion6 [30] (1,980 images, seven categories including neutral). we report the seven standard label-distribution metrics, which are Chebyshev, Clark, KL divergence (KLD), cosine similarity, Spearman rank correlation, the parameter-free approximate-correctness percentage μ [24], and the divisiveness error DVSE [25], which measures how much conflicting positive and negative emotion mass a prediction preserves and is therefore the metric closest to the ambiguity we target.

4.1.2. Implementation Details

Our default visual backbone is frozen CLIP ViT-L/14 [40]. The visual student is implemented as a lightweight residual MLP trained on top of a fixed SLR-C prior. In SLR-C, priors are constructed from three sources, namely lexical intent phrases, canonical intent descriptions, and Gemini 3.1 Pro-generated scenario descriptions [41] with frozen CLIP text encoder [40], and their normalized logits are averaged before local reranking within Top-10 visual candidates. For UTD, rationale text is generated offline by Qwen2.5-VL-7B-Instruct [42] and then encoded by a frozen bge-large-en-v1.5 text encoder [43]. The exact scenario- and rationale-generation prompt templates are released together with our artifacts. The teacher consumes only the resulting text embeddings and is optimized with Asymmetric Loss (ASL) [19] using $\gamma_- = 2$, $\gamma_+ = 0$, and clipping value 0.05. The student is optimized with AdamW [44], learning rate 5×10^{-4}, weight decay 10^{-4}, batch size 256, and dropout 0.1. The teacher's hidden dimension is 1,024, and the student's hidden dimension is 768. The distillation temperature is set to 2.0. Best-epoch selection and early stopping use validation class-wise Macro F1 only. All decision thresholds are fitted exclusively on the validation split and

Table 1: Main comparison on Intentonomy test split. Avg. is the mean of Macro/Micro/Samples F1. Rows under matched protocol (the "(ViT-L/14)" reproductions, the CLIP baseline, and ours) report the five-seed mean, and ± is the 95% *t*-interval half-width. †HVU-CLIP is quoted under its own ViT-B/32 backbone because no code has been released, and a backbone-matched comparison under ResNet101 is given in Table 13.

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
Random	-	6.94	7.18	7.10	7.07
Visual [31]	CVPR 2016	22.77	30.23	28.45	27.15
Multi-label classification methods					
HiMulConE [32]	CVPR 2022	28.43	41.50	42.15	37.36
MultiGuard [33]	NeurIPS 2022	26.87	38.67	40.06	35.20
SST [34]	AAAI 2022	27.33	42.21	42.62	37.39
Query2Label [35]	-	32.12	44.64	45.15	40.64
DualCoOp [36]	NeurIPS 2022	31.06	44.42	40.40	38.63
Label noise methods					
Cleannet [37]	CVPR 2018	25.89	42.15	40.97	36.34
LS [38]	ICML 2020	24.32	42.77	43.27	36.79
Dividemix [39]	ICLR 2020	30.11	40.23	41.56	37.30
Intent perception methods					
Jia [1]	CVPR 2021	23.98	31.28	31.39	28.88
CPAD [6]	TIP 2023	27.39	40.98	41.12	36.50
PIP-Net [5]	TMM 2023	32.57	45.94	47.05	41.85
CoT4Intent [12]	-	41.70	53.10	53.82	49.54
HVU-CLIP [10] (ViT-B/32)†	TPAMI 2026	42.91	55.22	52.54	50.22
LabCR (ViT-L/14) [8]	TIP 2024	$47.87_{\pm 0.38}$	$56.78_{\pm 0.97}$	$57.23_{\pm 1.25}$	$53.96_{\pm 0.71}$
HLEG (ViT-L/14) [7]	TAC 2023	$47.38_{\pm 0.75}$	$54.85_{\pm 1.70}$	$55.29_{\pm 1.06}$	$52.50_{\pm 0.98}$
IntCLIP (ViT-L/14) [9]	ECCV 2024	$44.98_{\pm 1.12}$	$54.73_{\pm 0.49}$	$54.73_{\pm 1.18}$	$51.48_{\pm 0.82}$
IntentMLM [11]	PR 2026	43.05	54.77	56.75	51.52
CLIP ViT-L/14 [40]	ICML 2021	$47.33_{\pm 1.15}$	$54.98_{\pm 2.90}$	$55.74_{\pm 2.64}$	$52.68_{\pm 1.96}$
FDIL (UTD only)	Ours	$49.90_{\pm 1.18}$	$58.01_{\pm 1.03}$	$58.43_{\pm 1.13}$	$55.45_{\pm 1.02}$
FDIL	Ours	$\mathbf{51.90}_{\pm 0.95}$	$\mathbf{59.90}_{\pm 0.62}$	$\mathbf{60.03}_{\pm 0.94}$	$\mathbf{57.28}_{\pm 0.75}$

then fixed before test evaluation, and test outputs are used solely for final reporting and never enter any model- or threshold-selection branch. Notably, UTD introduces *zero extra inference cost*, since the text teacher is only used during training and is discarded at test time.

4.2. Comparative Experiments

4.2.1. Comparison with state-of-the-art methods

Table 1 summarizes the main comparison on the test split. Under the matched protocol FDIL leads the baselines by +3.3 to +5.8 Avg. F1, each significant after correction on Avg. F1, mAP, and hard-subset F1. Against the strongest compared method, LabCR, the gain is +3.32 Avg. F1 (p = 0.002) and +7.67 hard-subset F1 (p = 0.004). UTD alone already improves Avg. F1 over the baseline at zero extra inference cost, and full FDIL improves all four metrics.

4.2.2. Zero-shot Multimodal Large Model Comparison

To address the concern that recent multimodal large language models may already solve visual intent recognition through prompting alone, we evaluate MLLMs zero-shot on the same Intentonomy label set, prompting each to return a probability over the candidate categories. As shown in Table 2, GPT-5.6 Sol substantially narrows the gap, since its Avg. F1 remains below the task-trained CLIP baseline and 8.24 below FDIL, whereas its threshold-free mAP and hard-subset F1 exceed both task-trained references. These mixed results show that a frontier general-purpose MLLM can provide highly competitive ranking and hard-class recognition without task-specific training, while dataset-specific learning still retains a clear advantage on aggregate fixed-threshold F1.

Table 2: Zero-shot MLLM comparison on the Intentonomy test split. ‡Published result quoted from IntentMLM [11].

Model	Macro	Micro	Samples	Avg.	mAP	Hard
Zero-shot prompting						
Qwen3.5-9B [45]	37.11	36.50	36.40	36.67	42.99	22.78
Step3-VL-10B [46]	38.21	40.21	36.10	38.17	42.58	26.64
Qwen3-VL-8B-Instruct [47]	38.51	40.75	39.63	39.63	38.84	28.37
InternVL3-8B [48]	37.54	41.85	40.50	39.97	37.67	25.51
Gemma-4-E4B-IT [49]	38.76	42.44	39.25	40.15	40.42	27.08
GPT-4o [50]‡	39.48	40.57	41.45	40.50	-	-
GPT-5.6 Sol [51]	47.58	49.64	49.91	49.04	**57.50**	**36.19**
Task-trained reference						
CLIP ViT-L/14 [40]	47.33	54.98	55.74	52.68	50.67	28.45
FDIL (ours)	**51.90**	**59.90**	**60.03**	**57.28**	55.10	35.02

4.2.3. Performance on Difficulty- and Content-Dependent Subsets

Table 3 reports intent subsets grouped by difficulty and by content dependency. On the easy subset FDIL trails Query2Label by 0.40 and HLEG by 0.30, a negligible gap on visually unambiguous categories where modelling semantic overlap offers little headroom. FDIL's largest and most consistent gains fall on the medium and hard subsets, exactly where semantic and supervisory ambiguity are most pronounced. On the content split, FDIL's major gain appears in the *Others* subset, where recognition requires holistic reasoning over multiple weak cues rather than a single object or scene pattern.

Table 3: Intent subsets grouped by difficulty and content dependency, where Avg. is the corresponding subset mean.

	Difficulty				Content dependency			
Method	Easy	Medium	Hard	Avg.	Object	Context	Others	Avg.
Random	19.86	7.11	2.81	9.93	7.75	12.53	6.05	8.78
Visual [31]	54.64	24.92	10.71	30.09	25.58	30.16	21.34	25.69
Jia (w/o hashtag) [1]	57.10	25.68	12.72	31.83	28.15	28.62	22.60	26.46
Jia (hashtag) [1]	58.86	26.30	13.11	32.76	29.66	32.48	22.61	28.25
Query2Label [35]	**81.59**	38.06	14.55	44.73	–	–	–	–
CPAD [6]	69.47	31.50	8.54	36.50	28.81	47.30	24.74	33.62
HLEG [7]	81.49	38.99	20.22	46.90	–	–	–	–
PIP-Net [5]	70.73	33.63	14.33	39.56	39.20	39.53	27.59	35.44
IntentMLM [11]	76.26	46.87	27.39	50.17	50.13	49.69	39.75	46.52
CLIP ViT-L/14 [40]	80.99	53.32	28.45	54.25	61.80	**61.69**	40.59	54.69
FDIL	81.19	**57.73**	**35.02**	**57.98**	**63.04**	61.32	**47.15**	**57.17**

4.2.4. Interpretability Analysis

Figure 3 shows a t-SNE projection of the decision-score space on four semantically adjacent intents. FDIL produces more compact, better-separated same-intent clusters than the CLIP baseline (silhouette $0.21 \rightarrow 0.29$), so the semantic prior and the rationale teacher pull confusable neighboring intents apart rather than merely rescaling scores. Per-class F1 changes under the matched class-wise protocol concentrate on hard and abstract classes, including *WorkILike* (+28.5), *PassionAboutSomething* (+17.3), and *SocialLifeFriendship* (+17.2), while only two classes regress.

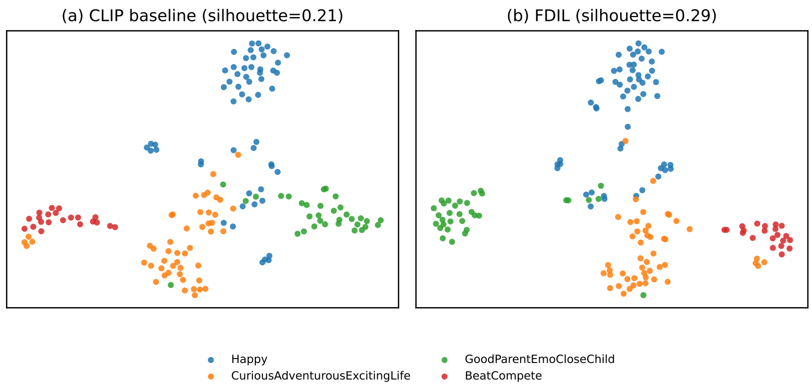

Figure 3: t-SNE of the decision-score space on four semantically adjacent intents. FDIL yields tighter, better-separated same-intent clusters than the CLIP baseline.

Table 4: Component ablation of FDIL, where *w/o UTD* removes text distillation and *Ungated teacher* removes agreement-aware gating.

Method	Macro	Micro	Samples	Avg.	Hard
w/o UTD	47.90	55.98	55.18	53.02	29.71
Ungated teacher	49.69	57.90	58.98	55.52	31.69
FDIL (full)	**52.13**	**60.06**	**60.28**	**57.49**	**35.02**

4.3. Ablation Studies

4.3.1. Effectiveness of SLR-C and Residual Refinement

Table 4 ablates SLR-C against residual learning and distillation. The prior alone is insufficient, as without a trainable student the mixed prior distorts the score geometry, and removing UTD costs 4.47 Avg. F1. Once distillation is present, residual refinement is consistently beneficial, so heterogeneous prior fusion and uncertainty-aware supervision must be optimized jointly rather than independently.

4.3.2. Effectiveness and Necessity of Functional Decoupling

Table 5 contrasts decoupled FDIL with three unified controls under identical frozen CLIP ViT-L/14 features, split, and validation-only class-wise thresholding. *Unified joint-target KD* averages the SLR-C and rationale-teacher probabilities into one distillation target and trains a single student, without residual decomposition or agreement-aware gating. Neither signal is degenerate in isolation, but the manner of combination is decisive.

Table 5: Unified versus functionally decoupled ambiguity handling. Results are means over three seeds.

Method	Macro	Micro	Samples	Avg.	Hard
Unified UTD only	50.20	58.03	58.79	55.68	30.27
Unified SLR-C only	51.29	**60.10**	59.22	56.87	34.01
Unified joint-target KD	50.69	58.35	57.96	55.66	32.41
FDIL	**52.13**	60.06	**60.28**	**57.49**	**35.02**

We further examine how the two modules respond to the two sources of ambiguity in Table 6. Supervisory ambiguity is measured by $1 - \omega$, with $\omega = 1/3$ defining the high-ambiguity stratum, while semantic ambiguity is measured by the negative top-2 margin of the out-of-fold baseline logits and split at the median. Significance is assessed using paired two-sided t-tests on per-sample F1 against the shared baseline, with Holm correction over the twelve comparisons. The results reveal different sensitivity patterns for the two modules. SLR-C consistently benefits samples with high semantic ambiguity, improving F1 by +4.08 and +1.17 under low and high supervisory ambiguity, respectively, with both gains remaining significant after correction. In contrast, UTD has essentially no effect when annotator agreement is high (−0.05 and −0.07 F1), while its largest and only significant gain appears when high supervisory ambiguity coincides with high semantic ambiguity (+0.84, corrected $p = 0.025$). Thus, SLR-C primarily addresses ambiguity in the label semantics, whereas the benefit of UTD emerges under uncertain supervision and becomes most evident on semantically difficult samples. These complementary response patterns provide empirical support for assigning the two ambiguity sources to separate mechanisms.

Table 6: 2×2 ambiguity interaction on cross-fitted Intentonomy train predictions. Cells give Δ per-sample F1 against the shared baseline, and * marks Holm-corrected $p < 0.05$.

		Low supervisory ($\omega \geq 2/3$)		High supervisory ($\omega = 1/3$)	
		Low sem.	High sem.	Low sem.	High sem.
	n	1,271	588	5,099	5,781
Base	F1	72.23	41.56	40.54	22.90
ΔF1	+SLR-C	−0.43	+4.08*	+0.49	+1.17*
	+UTD	−0.05	−0.07	+0.03	+0.84*
	Full FDIL	−0.71	+2.18	+0.15	+1.14*

4.3.3. *Heterogeneous Priors in SLR-C*

Table 7 compares the three semantic sources of SLR-C under the reranking stage only, namely *Lexical* (short class-name phrases), *Canonical* (the official Intentonomy intent descriptions), and *Scenario* (Gemini-generated visual scenarios).

Table 7: Ablation of textual priors in SLR-C (reranking-only, K=10). Results use a single seed.

Lexical	Canonical	Scenario	Macro	Micro	Samples	Avg.	Hard
✓			49.76	59.12	**58.60**	55.83	31.22
	✓		49.80	59.50	58.05	55.78	29.88
		✓	49.35	58.88	57.87	55.37	32.48
✓	✓		**51.18**	59.39	58.11	56.23	32.92
✓		✓	50.12	59.30	58.36	55.93	32.75
	✓	✓	50.93	59.51	58.18	56.21	**34.81**
✓	✓	✓	50.90	**59.81**	58.47	**56.40**	32.80

No single source dominates, so the choice of prior is a trade-off rather than a strict ordering, whereas the three-source ensemble is best on the aggregate metrics and carries the broadest semantic coverage. The sources are therefore complementary, and we adopt the ensemble in FDIL, where it also gives the strongest hard-subset result under the full pipeline.

4.3.4. *Sensitivity to Candidate Size K*

Table 8 retrains FDIL under $K \in \{5, 10, 28\}$ with class-wise thresholding. $K = 10$ is best overall and is our default, while $K = 5$ is slightly weaker and $K = 28$ comparable but no better. A moderate candidate set therefore balances semantic coverage against the distractor noise of an overly broad reranking range.

Table 8: Sensitivity to candidate size K (class-wise thresholding). Results are means over three seeds.

Top-K	Macro	Micro	Samples	Avg.	Hard
5	51.19	59.03	59.42	56.54	31.60
10	**52.13**	**60.06**	**60.28**	**57.49**	**35.02**
28	51.64	59.59	59.71	56.98	34.02

Table 9: Calibration decomposition under matched global/class-wise thresholding. FDIL is mean±std over three seeds, and CLIP uses a single seed.

Method	Threshold	Macro	Micro	Samples	Avg.	mAP	Hard
CLIP ViT-L/14 [40]	Global	46.22	55.05	54.51	51.92	50.67	26.84
CLIP ViT-L/14 [40]	Class-wise	47.40	56.42	56.41	53.41	50.67	28.45
FDIL	Global	$47.70_{\pm 1.55}$	$57.56_{\pm 0.41}$	$57.43_{\pm 0.69}$	$54.23_{\pm 0.72}$	$\mathbf{55.10}_{\pm 0.26}$	$30.00_{\pm 3.46}$
FDIL	Class-wise	$\mathbf{52.13}_{\pm 0.81}$	$\mathbf{60.06}_{\pm 0.24}$	$\mathbf{60.28}_{\pm 0.42}$	$\mathbf{57.49}_{\pm 0.49}$	$\mathbf{55.10}_{\pm 0.26}$	$\mathbf{35.02}_{\pm 1.77}$

4.3.5. Calibration and Threshold-free Evaluation

Table 9 evaluates the baseline and FDIL under matched global and class-wise thresholding. FDIL improves on average and hard-subset F1 under both protocols, and class-wise thresholding helps both models, so calibration is a shared post-processing gain rather than the source of FDIL's advantage, and the decision-rule-invariant mAP confirms this with a +4.4 gain.

Table 10: Threshold-free, rank-based metrics. mAP is the macro average precision. FDIL/UTD mAP values are means over three seeds, and all AUCs and CLIP mAP use a single representative seed.

Method	mAP	Macro ROC-AUC	Micro ROC-AUC	Micro PR-AUC
CLIP ViT-L/14 [40]	50.67	87.04	84.90	54.89
UTD only	53.07	**90.09**	**88.55**	59.59
SLR-C only	53.79	88.24	86.73	58.60
FDIL	**55.10**	89.36	88.31	**60.64**

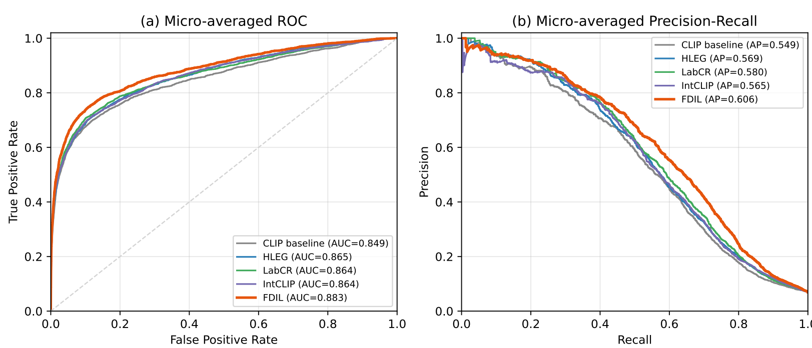

Figure 4: Micro-averaged ROC (a) and precision-recall (b) curves, comparing FDIL against the CLIP baseline and three representative methods (HLEG, LabCR, IntCLIP) reproduced under the matched CLIP ViT-L/14 features, split, and protocol.

Table 10 and Figure 4 report threshold-free, rank-based metrics. FDIL improves over the baseline on every metric with a 95% paired-bootstrap interval excluding zero ($p < 0.001$), and leads the reproduced methods once the backbone is equalized.

4.3.6. The Effectiveness of Rationale

Table 11 ablates the teacher text under gated distillation without SLR-C, comparing a plain Caption against rationales built from observable visual evidence (Step 1), contextual interpretation (Step 2), and counterfactual disambiguation (Step 3).

Table 11: Ablation of rationale construction for UTD, where Steps 1, 2, and 3 add visual evidence, contextual bridging, and counterfactual disambiguation. Results use a single seed.

Method	Macro	Micro	Samples	Avg.	Hard
Caption	49.64	**58.64**	58.29	55.52	29.14
Rationale (Step 1)	49.81	56.39	57.65	54.62	29.28
Rationale (Step 1 & 2)	49.69	54.45	55.97	53.37	28.82
Rationale (Step 1 & 2 & 3)	**50.31**	57.97	**59.07**	**55.78**	**30.24**

Caption supervision is already a reasonably strong privileged signal, yet it trails the full rationale on Macro, Samples, Avg., and Hard. Adding contextual interpretation alone (Step 2) slightly degrades the metrics, indicating semantic drift without explicit disambiguation, and the counterfactual Step 3 recovers and surpasses it. Subjective intent supervision therefore requires not only richer context but explicit boundary contraction among confusing alternatives.

4.3.7. Negative Controls on the Text Teacher

Table 12 retrains FDIL with a single ingredient corrupted, using shuffled rationale (each image paired with another image's rationale), ungated teacher (standard KD), and uniform gate (every sample at the mean agreement), against the supervised residual (w/o UTD) as the lower reference. Shuffling rationales degrades more than removing UTD altogether (-10.40 against -4.47 Avg. F1), so the gain relies on image-specific rationale content rather than on arbitrary external text. Flattening or removing the agreement gate degrades less but consistently, supporting uncertainty-aware gating.

Table 12: Negative controls on the UTD text teacher, with Δ in points, where negative denotes degradation.

Condition	Macro	Micro	Samples	Avg.	mAP	Hard
FDIL (full)	52.13	60.06	60.28	57.49	55.10	35.02
w/o UTD	−4.23	−4.08	−5.10	−4.47	−4.93	−5.31
Shuffled rationale	−7.75	−12.39	−11.01	−10.40	−9.04	−7.58
Ungated teacher	−2.44	−2.16	−1.30	−1.97	−1.49	−3.33
Uniform agreement gate	−1.16	−0.93	−1.21	−1.12	−1.33	+1.13

4.4. Backbone Independence

Table 13 compares the performance of FDIL and strong existing methods under ResNet101. FDIL remains clearly effective on ResNet101, showing that the decoupled design is not exclusive to CLIP features. Compared with HVU-CLIP, FDIL demonstrates superior performance on the Macro, Samples, and Avg. metrics, which could be attributed to its ability to handle the hard categories with high ambiguity.

Table 13: Comparison of models under ResNet101 on the Intentonomy test split.

Method	Macro	Micro	Samples	Avg.	Hard
Baseline	36.56	41.87	43.10	40.51	18.99
IntCLIP [9]	40.15	**53.54**	51.01	48.23	-
HVU-CLIP [10]	41.24	52.22	50.41	47.96	-
FDIL	**43.20**	50.63	**52.02**	**48.62**	**25.65**

4.5. Cross-Dataset Generalization

We further evaluate FDIL under label distribution learning (LDL), where the target is a probability distribution over labels rather than a set of binary decisions. LDL requires no threshold selection, so it tests whether the functional decoupling generalizes beyond the prediction space it was developed for. Table 14 reports both datasets.

The reproduced LDL objective LDL-DVS [25] performs close to the matched baseline in this setting. This suggests that the stronger pretrained representation substantially reduces the optimization headroom available to these regularizers compared with the visual features used in their original evaluations. The two FDIL components exhibit complementary effects across the two datasets. SLR-C mainly improves distribution-fitting metrics (KLD, cosine similarity, and μ) on Flickr-LDL, and similarly improves

Table 14: Results on Flickr-LDL and Emotion6, averaged over five seeds. */** indicate significant improvements over the matched baseline at $p < 0.05/p < 0.01$, and † indicates a significant degradation at $p < 0.05$ (paired t-test over five paired runs). Bold and underline mark the best and second-best entry among the baseline and the three FDIL variants.

Method	Cheby.↓	Clark↓	KLD↓	Cos.↑	Spear.↑	μ (%)↑	DVSE↓
Flickr-LDL (2,229 test images)							
LDL-DVS [25]	.20385	2.30341	.34332	.89663	.71085	71.40	.07719
Baseline	.20386	<u>2.30331</u>	<u>.34329</u>	.89664	.71089	71.40	.07723
+ SLR-C	.20308	**2.30313**	**.34061**	**.89732**	.71291	**71.58**	.07641
+ UTD	**.20257**	2.31226†	.34542	.89610	<u>.71369</u>	71.31	**.07292***
FDIL	<u>.20272</u>	2.31126†	.34332	<u>.89672</u>	**.71406***	<u>71.45</u>	<u>.07379**</u>
Emotion6 (594 test images)							
LDL-DVS [25]	.17939	1.59807	.25322	.89464	.72225	64.20	.07834
Baseline	**.17937**	1.59790	.25317	.89466	.72227	64.20	.07838
+ SLR-C	<u>.17968</u>	1.59758	<u>.25152</u>	**.89548**	<u>.72314</u>	**64.52**	**.07811**
+ UTD	.18044	<u>1.59491</u>	.25372	.89406	**.72563***	64.12	.07907
FDIL	.18017	**1.59097***	**.25132**	<u>.89528</u>	.72194	<u>64.41</u>	<u>.07812</u>

these metrics on Emotion6. UTD has a stronger effect on distribution structure, improving Spearman correlation on both datasets and reaching statistical significance on Emotion6 ($p = 0.042$). On Flickr-LDL, it also significantly reduces DVSE ($p = 0.010$), although this is accompanied by a degradation in Clark distance. These results are consistent with the distinct roles of the two components that SLR-C refines the allocation of probability mass among semantically plausible labels, whereas UTD exploits uncertainty in the target distribution to guide the learned prediction structure. Two caveats apply. The margins are small, FDIL significantly degrades Clark distance on Flickr-LDL, and the two datasets disagree on which metric improves. We therefore read these results as a transfer test of the functional decoupling, whose division of labour replicates on both datasets, rather than as a new state of the art for LDL.

4.6. *Efficiency Comparison*

Table 15 compares the inference efficiency of representative methods. FDIL (UTD only) has exactly the same inference cost as the baseline because the text teacher is discarded after training. The full FDIL requires only 2.41M parameters and 4.93M FLOPs, with a latency of 0.466 ms per image. In comparison, HLEG introduces sub-

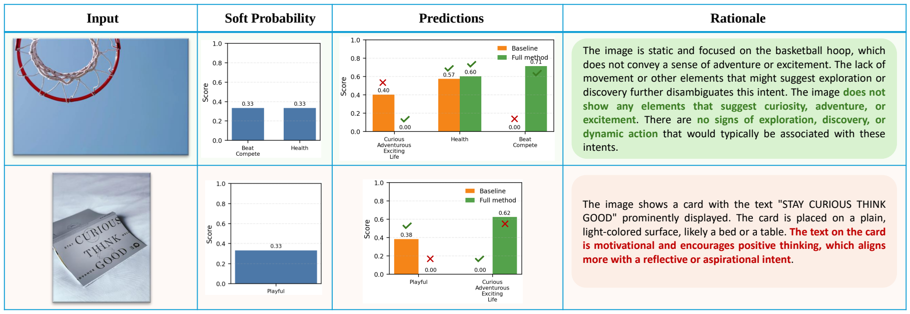

Figure 5: Qualitative case study of baseline and FDIL on representative Intentonomy samples.

stantially heavier Transformer decoding. IntCLIP is the cheapest head measured here, but its region-aggregation design still runs on top of the same frozen encoder and trails FDIL by 5.80 Avg. F1 under the matched ViT-L/14 protocol of Table 1. HVU-CLIP releases no code, so it cannot be measured directly, and we expect a cost comparable to IntCLIP. IntentMLM and CoT4Intent follow MLLM inference pipelines, whose cost is orders of magnitude higher. Overall, FDIL achieves its gains with only minor inference overhead.

Table 15: Inference efficiency comparison on an RTX 4090.

Method	Params (M)	FLOPs (M)	Latency (ms)
Baseline	1.20	2.40	0.109
HLEG [7]	203.70	12 075.9	2.725
IntCLIP [9]	0.02	22.02	0.111
FDIL (UTD only)	1.20	2.40	0.109
FDIL	2.41	4.93	0.466

4.7. Case Study

Figure 5 provides qualitative examples. In the first, FDIL suppresses a plausible but incorrect distractor using counterfactual rationale evidence, consistent with UTD sharpening semantic boundaries. In the second, FDIL errs on a label with only 0.33 agreement yet stays semantically plausible, illustrating the intrinsic uncertainty of subjective intent recognition.

5. Discussion

FDIL rests on the two ambiguities of intent recognition. Keeping them separate makes each function independently justified, measurable, and deployable. FDIL is suitable for offline or high-throughput visual-content analysis, because its external semantic knowledge can be prepared offline. UTD incurs no additional inference cost, as the rationale generator and text teacher are discarded after training, while SLR-C only requires lightweight prior-based reranking and residual refinement.

Despite the consistent gains, the limitations lie in prior and visual representation. Offline text generation and prior construction add preparation cost. Gains shrink on weaker backbones such as ResNet101, so residual refinement still relies on reasonably strong base representations, and richer priors are not automatically better unless their interaction with the student is controlled. For a few broadly defined, overlapping categories the residual student can resolve the candidate set toward a competing neighbor and lose accuracy on the original class. Beyond the multi-label setting the transferred gains are not uniform across metrics. On the label-distribution benchmarks FDIL improves rank correlation and divisiveness on Flickr-LDL but degrades Clark distance there, and the two datasets do not agree on which metric improves. Natural next steps are source-aware prior weighting instead of uniform averaging, stronger students for weaker backbones, and confusion-pair-specific textual evidence rather than only global distillation and local reranking.

6. Conclusion

In this work, we study subjective visual intention recognition under supervisory ambiguity and semantic ambiguity, and propose FDIL as a functionally decoupled framework. In the current formulation, fixed semantic priors provide structured candidate semantics, a residual student supports image-aware intent prediction on top of the semantic prior, and uncertainty-aware text teachers stabilize ambiguous supervision.

Extensive experiments on Intentonomy show that the proposed framework consistently improves over strong visual baselines and prior state-of-the-art methods. Cross-dataset studies on the Flickr-LDL and Emotion6 label-distribution benchmarks show that the division of labour between the two modules persists, with the semantic prior moving pointwise distribution fit and the uncertainty-guided teacher moving rank correlation. Overall, the results suggest that once visual representations are sufficiently strong, the key challenge of intention understanding lies in resolving supervisory and semantic ambiguity rather than in extracting low-level visual features alone. We hope this functionally decoupled formulation provides a clearer perspective for future research on human-centered visual understanding tasks.

Data and artifact availability

The Intentonomy, Flickr-LDL, and Emotion6 benchmarks are all publicly available. Upon acceptance, we will release generated priors, rationales, cached features/logits, prompts, thresholds, seeds, checksums, and training/evaluation scripts.

CRediT authorship contribution statement

Yin Tang: Writing - original draft, Validation, Formal analysis; **Lifeng Fan:** Supervision, Writing - original draft, Conceptualization; **Hongyu Yang:** Writing - original draft, Formal analysis; **Xuan Dong:** Writing - original draft, Formal analysis; **Weixin Li:** Writing - original draft, Supervision, Resources, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT in order to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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